photosynthesis) is converted to sucrose. Sucrose moves into the companion cells, then into the living phloem sieve tube cells by active transport. This process produces a hypertonic condition in the phloem. Water in the adjacent xylem moves into the phloem by osmosis. As osmotic pressure builds up, the phloem sap moves to areas of lower pressure. By active transport sucrose moves into the cells where it is utilised or stored. As sugars are removed, the osmotic pressure decreases and water moves out of the phloem.

14.11 Ascent of Sap and its Events – An Overview

The upward movement of water and minerals from roots to different plant parts is called ascent of sap. A number of factors play a role in ascent of sap and it takes places in following steps

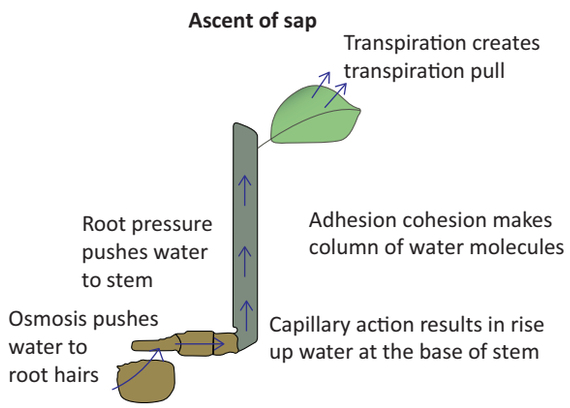


Figure 14.8 Ascent of Sap

1. **Root Pressure:** Water from soil enters the root hairs due to osmosis. Root pressure is responsible for movement of water up to the base of the stem.
2. **Capillary Action:** Water or any liquid rises in a capillary tube because of physical forces, this phenomenon is called capillary action. In the same way, in stem water rises up to certain height because of capillary action.
3. **Adhesion-cohesion of Water Molecules:** Water molecules form a continuous column in the xylem because of forces of adhesion and cohesion among the molecules.
4. **Cohesion:** The force of attraction between molecules of water is called cohesion.
5. **Adhesion:** The force of attraction between molecules of different substances is called adhesion.Water molecules stick to a xylem because of force of adhesion.

More to Know

Dews like water droplets on the leaves of grass seen in the early mornings, when the climate is humid and excess of water is present in the plants, the excess water is exudated in the form of liquid. This is due to root pressure .This phenomenon is called **Guttation** which takes place through specialized cells called **Hydathodes**.

Transpiration Pull: Transpiration through stomata creates vacuum which creates a suction. called transpiration pull. The transpiration pull sucks the water column from the xylem tubes and thus water is able to rise to great heights even in the tallest plants.

Activity 2

Demonstration of Root Pressure

Choose a small soft stemed plant. Cut the stem horizontally near the base with a blade in the morning. You will see drops of solution oozing out of the cut stem due to root pressure.

14.12 Blood

Blood is the main circulatory medium in the human body. It is a red coloured fluid connective tissue.

Components of Blood: The blood consists of two main components. The fluid **plasma** and the **formed elements** (blood cells) which are found suspended in the plasma.

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